Assignment No. 10

Title: Practical Based on Implementing Date and Time Functions

Objective:

To understand and apply various **Date and Time functions** in SQL Server using the **AdventureWorks** database.

Theory:

Date and Time functions are used to manipulate, extract, and calculate values related to dates and times.

They are commonly used for generating reports, tracking sales periods, and computing durations.

Commonly used Date & Time functions in SQL Server:

Function	Description
GETDATE()	Returns the current system date and time
SYSDATETIME()	Returns current date and time with more precision
YEAR(date)	Extracts the year part of a date
MONTH(date)	Extracts the month part of a date
DAY(date)	Extracts the day part of a date
DATEPART(part, date)	Returns a specific part of a date (year, month, day, weekday, etc.)
DATENAME(part, date)	Returns the name of a specific date part
DATEADD(part, number, date)	Adds a specified number of units to a date
DATEDIFF(part, start, end)	Returns the difference between two dates
EOMONTH(date)	Returns the last day of the month for a given date
FORMAT(date, format)	Formats a date in a specified style

Table Used:

Sales.SalesOrderHeader

Column Name	Data Type	Description
SalesOrderID	INT	Unique Order ID
OrderDate	DATETIME	Date when order was placed
DueDate	DATETIME	Due date for order
ShipDate	DATETIME	Date when order was shipped

Queries:

1) Display the current system date and time.

```
SELECT GETDATE() AS CurrentDateTime;
```

2) Display the year, month, and day from the current date.

```
SELECT
   YEAR(GETDATE()) AS CurrentYear,
   MONTH(GETDATE()) AS CurrentMonth,
   DAY(GETDATE()) AS CurrentDay;
```

3) Display OrderDate along with extracted Year and Month.

```
SELECT
   SalesOrderID,
   OrderDate,
   YEAR(OrderDate) AS OrderYear,
   MONTH(OrderDate) AS OrderMonth
FROM Sales.SalesOrderHeader;
```

4) Find the number of days between OrderDate and ShipDate.

```
SELECT
   SalesOrderID,
   OrderDate,
   ShipDate,
   DATEDIFF(DAY, OrderDate, ShipDate) AS DaysToShip
FROM Sales.SalesOrderHeader;
```

5) Add 10 days to each OrderDate.

```
SELECT
   SalesOrderID,
   OrderDate,
   DATEADD(DAY, 10, OrderDate) AS NewDate
FROM Sales.SalesOrderHeader;
```

6) Display the last day of the month for each OrderDate.

```
SELECT
   SalesOrderID,
   OrderDate,
   EOMONTH(OrderDate) AS EndOfMonth
FROM Sales.SalesOrderHeader;
```

7) Display the name of the day for each OrderDate.

```
SELECT
   SalesOrderID,
   OrderDate,
   DATENAME(WEEKDAY, OrderDate) AS DayName
FROM Sales.SalesOrderHeader;
```

8) Display OrderDate in a formatted style (DD-MMM-YYYY).

```
SELECT
   OrderDate,
   FORMAT(OrderDate, 'dd-MMM-yyyy') AS FormattedDate
FROM Sales.SalesOrderHeader;
```

9) Find total number of years between the earliest and latest OrderDate.

```
SELECT
   DATEDIFF(YEAR, MIN(OrderDate), MAX(OrderDate)) AS TotalYears
FROM Sales.SalesOrderHeader;
```

10) Display the current system date with higher precision using SYSDATETIME().

```
SELECT SYSDATETIME() AS PreciseDateTime;
```

Output:

Results display transformed date and time values based on each function used.

Conclusion:

Date and Time functions in SQL Server are highly useful for performing calculations, formatting, and extracting specific components of date/time data. They are essential in reporting and analytics applications.